



Collaboration patterns and patenting: Exploring gender distinctions



Yu Meng*

Public Policy and Global Affairs Programme, Nanyang Technological University & Longus Institute for Development and Strategy, Singapore

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ABSTRACT

Drawing upon the theory of gender frame, the research on gender in science, and social network studies, this paper focuses on the social mechanism of collaboration, specifically the *boundary-spanning collaboration*, to understand the gender gap in academic patenting in the U.S. Correspondingly, the author developed a few hypotheses for empirical testing. The results show that, else being equal, only collaboration with industry would significantly increase the probability of patenting for female academic scientists, but this helps explain considerable difference in patenting between female and male academics. The findings are discussed along with the limitations and policy implications at the end.

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1. Introduction

Both female doctoral degree recipients in science and engineering (S&E) and female professionals in scientific workplace have increased steadily in recent decades (Rossiter, 1995; NSF, 2008), but research has continuously found sexism against women in science (Editorial, 2013). The persistent and pervasive gender inequalities in science are detrimental to social values and economic development by reinforcing negative stereotypes and discouraging talented people from participating in and contributing to science (Hanson, 1996; Fox, 2008). Economically, it is estimated by Hunt et al. (2012) that closing the gap between female and male S&E degree holders would increase gross domestic product (GDP) per capita by 2.7% in the U.S. Other socially harmful consequences include the loss of talents and the exclusion of the specific types of knowledge women develop and maintain (Schiebinger, 2008; Kugele, 2010). Hence, the gender gap in science is deemed an ongoing focus of scholarly research and policy intervention.

Research productivity, because of its central role in the success of a scientific career, has attracted the most attention (Fox and Stephan, 2001). While vast previous research on gender and productivity focused on publication records and documented a pattern favoring male scientists (see Cole and Zuckerman, 1984; Xie and Shauman, 1998; Pripic, 2002 for literature review),

concerns have been increasingly cast on another indication of research productivity, *academic entrepreneurship*¹ (Ding and Choi, 2011; Franzoni and Lissoni, 2009) or academic research targeting commercial returns through patents, licenses, products, and many other forms of outcomes. Since commercial involvement is assumed to closely related to established status and a high level of publication productivity (Zucker and Darby, 1996; Stuart and Ding, 2006) and women are generally less successful on these aspects, scholars worry that the change toward commercialization would reinforce women's disadvantages in the profession of science (Whittington and Smith-Doerr, 2005, 2008; Ding et al., 2006; Murray and Graham, 2007; Haeussler and Colyvas, 2011).

Correspondingly, scholars have set out to examine women's status in various forms of academic entrepreneurship, from disclosure (Duque et al., 2005), licensing (Thursby and Thursby, 2005), serving in a firm's scientific advisory board or SAB (Stephan and El-Ganainy, 2007), patenting (e.g., Whittington and Smith-Doerr, 2005; Frietsch et al., 2009; Meng and Shapira, 2010; Blume-Kohout, 2014), to firm founding (Ding, 2004). These efforts have resulted in a consistent finding of a gender gap to women's disadvantage, but have not yet reached to the consensus regarding the underlying explanations. Related to the ambiguity is insufficient attention to collaboration, a process generally assumed to promote academics' research and commercial productivity (see Bozeman et al., 2013

¹ Some scholars use this concept to refer only to firm founding (e.g. Ding and Choi, 2011), but I adopted a broader definition that refers to various commercial activities including patenting, consulting, research collaboration with industry, and firm formation (Franzoni and Lissoni, 2009).

* Tel.: +65 91722579.

E-mail address: mengyu2211@hotmail.com

for the review) and be responsible for women's lower productivity (Kyvik and Teigen, 1996). Attempting to fill the research gaps, this study investigates the gender distinctions in academic patenting with a focus on academics' collaboration ties.

2. Literature review

2.1. The foundation: the theory of gender frame

Gender is one of a very few primary frames in our society that guide the organization of social practices (Ridgeway, 1997, 2007). According to Ridgeway, people have to develop “common” knowledge as a basis for their interaction and coordination with others in everyday life. How to categorize self and others is a piece of common knowledge that is useful to define the situation and make sense of one another in social practices. Meanwhile, such a category system should be simplified to allow for real-time management of actions. Eventually, the developmental process results in only a few cultural categories (gender, race, class, religion, etc.) but they serve as the primary guidance for individual perception and action in relation to others (Brewer and Liu, 1989; Ridgeway, 2006, 2007, 2009).

As gender is used as a primary cultural frame for differentiation and categorization, “difference is easily transformed into inequality through any of a variety of social processes” despite “difference need not logically imply inequality” (Ridgeway, 2009, p. 149). However, during the course of pursuing mutual dependence of groups for societal survival and stabilization, agreement gradually emerged among members across groups on which group is more respected and status-worthy than others (Ridgeway, 2006). While consensual gender beliefs (or *stereotypes* that are held by both men and women) are evidenced, research also found that these beliefs view men to be more proactive and competent in general and especially highly competent at the things that “count most” in society; and view women to be less competent generally but better at more feminine, communal tasks that tend to be socially less valued (see Ridgeway, 2006 for a comprehensive review). For instance, psychological studies (Hilton and von Hippel, 1996; Barbercheck, 2001; Powell et al., 2002) have shown that both sexes tend to consider men are assuming agentive qualities (being assertive, competitive, aggressive, courageous, instrumental, etc.) but not women. The association of sexes with differential attributes in the beliefs operates to women's disadvantage in the workforce, preventing them from participating in work attached with the highest social value.

Like other cultural beliefs, the gender frame endures over time (Heilman et al., 1989; Powell et al., 2002; Tinkler et al., 2015). It is reinforced through cognitive mechanisms and socialization. Even though the frame may change in the presence of disconfirming information, deviants, or new exemplars, people are likely to maintain it than to change it (see the review in Powell et al., 2002). Additionally, the belief is powerful in that even people consciously supporting gender equality would behave in biased ways just because they perceive *others* are influenced by such beliefs (Motowidlo, 1986; Dovidio et al., 1988). Together, the research suggests the gender frame has strong and enduring effects in the contemporary society and works against women's participation and progress in domains that are highly socially valuable such as science.

2.2. Gender and scientific productivity: traditional research and proposed explanations

Scientists work in all workforce sectors, but doctoral-level female scientists tend to work in educational (especially higher

education) institutions (Fox, 2001). Given this context, prior research concerning gender inequalities in science primarily focused on academics and their publication productivity. As early as forty years ago, scholars already reported the general lower level of women's publication productivity (Cole, 1979) and, since then, this result has been echoed in a vast number of studies and deemed a puzzle attracting huge efforts. It was claimed recently that “most of the observed sex differences in research productivity can be attributed to sex differences in personal characteristics, structural position and marital status” (Xie and Shauman, 1998, p. 847). But the authors reminded us the puzzle has not been completely resolved as the effects of gender-specific attributes and social processes are not deciphered yet. While these two categories of factors work together to produce disparities, we should note they are not equally important: individual attributes (including demographic characteristics and education background) may explain part of women's disadvantages, but they do not exist in a vacuum but are shaped by social and structural factors (Fox, 1991). In this sense, more attention should be devoted to the social processes and the way they work to differentiate women's locations and access to strategic resources. Then the theory of gender frame, together with some other theoretical explanations, could help us understand how social processes work fundamentally and differently for the sexes in science.

Science closely connects with power (Fox, 2001). As Fox articulated, science not only has critical consequences for human's living but links to powerful social institutions (especially the state and education). Additionally, science has been imprinted as a masculine area as it has long been dominated by men with women's entry being a recent phenomenon (Rossiter, 1982, 1995). Given its close connection with power and its history, it is unsurprising that science would see rather strong gender beliefs that run against women. Indeed, prior research has revealed that women are generally treated in science as “token” (Kanter, 1977a,b), “outsider” (Zuckerman et al., 1991), “stranger” (Sonnert and Holton, 1995a), and “illegitimate group members” (Burt, 1998).

As stated earlier, these beliefs are reinforced by various social processes and thus become further resistant to changes. This is especially true in science. The production of scientific knowledge is a complex process involving many interactions among scientists, like discussing for problem identification, exchanging information for idea generation, and debating experimental design and data interpretation. In these various interactions, men as incumbents and decision-makers tend to select other men to interact and work together, revealing the “similar-to-me” effect that Kanter and other scholars discovered in top management (Rand and Wexley, 1975; Kanter, 1977a; Pulakos and Wexley, 1983). With these arguments in mind, we would expect that women are excluded and thus disadvantaged in almost all interactions of importance to research productivity. However, this does not mean we should regard the various interactions/processes equally or there is no need to investigate in detail how the gender differences emerge in a particular process. On the contrary, these issues are of impelling demand to understand and then convert the currently stalled progress toward gender equality in science. In addition, no matter how hard it is to eliminate the gender stereotypes, the possibilities exist that they may change or the actual effects may vary in specific contexts (Rothbart, 1981; Weber and Crocker, 1983; Ridgeway, 2009). Motivated by the call and the possibility, I inquire (1) what would be the most relevant social process, (2) how the gender beliefs would be introduced to the process, and (3) how the gender beliefs would affect the ultimate productivity level through the process, and let these inquiries lead my further search in the literature.

2.3. Academic patenting and gender

Gender differences in different forms of academic entrepreneurship have been studied, but patenting received the most frequent and more sophisticated investigations for conceptual and practical advantages (see Meng's dissertation for discussion, p. 3–4). The research on gender and inventive and intellectual property right pursuing activities is not a brand new endeavor, but earlier studies tended to provide anecdotal or historical accounts of women inventors and their inventions (e.g. Vare, 1988; Macdonald, 1992; Stanley, 1993). Only until recent few years have empirical and systematic investigations of the gender pattern of patenting emerged. And in the recent efforts, two distinct approaches exist although the results yielded all point to a gender gap to women's disadvantage.

One approach, initially developed by Naldi and Parenti (2002) and adopted by several other researchers, purely depends on the information from patent documents stored in different national or international patent offices. This approach, while having an advantage in showing the general pattern across workforce sectors, has embedded limitations. First, patent data do not allow studying those who have never patented, preventing the acquirement of more meaningful and more precise rate of patenting by gender. Second, patent documents do not record detailed information about inventors' demographic background, career history, network features, and employer's characteristics, preventing more sophisticated analyses and a more comprehensive understanding.

The other approach relying on survey data, usually combined with patent record information, has more potential in developing a theoretical understanding of the gender differences in patenting. For instance, Whittington and Smith-Doerr (2008) claimed that organizational structure is a critical variable to account for the gap and argued a networking organizational structure would be conducive to more patenting engagement for women. Ding et al. (2006) argued, based on their analyses of the information collected from academic respondents, the lack of exposure to commercial sector and suspicion of the rewards of commercial engagement hinder women's patenting activities. Murray and Graham (2007) found the gender difference was larger in older cohorts and argued it was the opportunity structure created by supply and demand factors as well as their interactions that placed female scientists in a disadvantaged situation in patenting and other entrepreneurial activities. Similarly, Stephan and El-Ganainy (2007) also provided a variety of explanations grouped as demand and supply factors for women's lower level of participation in patenting and other entrepreneurial activities.

No evidence has suggested that women are less capable of patenting their research (Swede, 2003). Indeed, the explanations reviewed above implies that the gap tends to result from a variety of social processes in science underlying which the gender beliefs play a substantial and active role. Moreover, an additional point could be made from the review is that lacking important social ties places women in disadvantages in accessing resources such as information, knowledge about commercial processes, decision-making authority, and so forth. Therefore, despite the gap is generally expected, more attention could be devoted to academics' social ties to understand how they work differentially for women and men to contribute to the gap and how their working logic could be changed to narrow the gap.

2.4. Relating collaboration to gender and patenting from the perspective of social ties

It is generally assumed that collaboration can enhance scientific productivity (Price and Beaver, 1966; Gordon, 1980; Walsh and Maloney, 2002 among many others). However, to what extent this relationship can be generalized is still questionable. There

exists evidence of no direct association (e.g. Duque et al., 2005) and critiques on the research method upon which the conclusion of positive association is based (Katz and Martin, 1997; Bozeman and Corley, 2004). When gender is further considered, mixing arguments and evidence (Kyvik and Teigen, 1996; Welch and Melkers, 2006; Hunter and Leahey, 2008; Bozeman and Gaughan, 2011) bring more challenges to clarifying the association.

In the attempt to detect the actual effects of collaboration on productivity, the social capital theory and network analysis provide a very useful perspective. As individuals are connected by social relationships (*ties*) of various kinds, for each kind of relationships they become embedded in an entity (*network*) that has structural features and poses social influence on them. This point has been well taken in social network and social capital theory (Marin and Wellman, 2010). In the social studies of science, although not specifically targeting collaboration ties, scholars have increasingly examined the relationship between scientists' social ties and their performance in creating knowledge and technology (e.g. Allen, 1977; Tushman, 1977; Walsh and Maloney, 2002; Rost, 2011). The social network perspective is taken here also because it could be quite useful to understanding the gender pattern in patenting as pointed out before.

Before further discussion, we should bear in mind, regardless of its increasing importance, patenting to faculty scientists is by and large an "optional" activity and involves costs. As Meng et al. (2013) noted, to apply for a patent, academic scientists have to devote additional time at least to a great deal of paper work around patent filing, if not to other responsibilities (e.g. finding a patent lawyer, seeking for a potential industrial licensee, etc.). In addition, as universities traditionally function as a noncommercial environment, faculty scientists are likely to have little knowledge about entrepreneurial process and face difficulties in terms of university support (Mowery, 2007). As a result, time has to be reallocated between traditional academic responsibilities and commercial activities. Lack of balance between these two broad categories of activities tends to hurt the academic career progress. Also, research found that technologies demanded by the market were more applicable and do not reflect academic research frontier (Trajtenberg et al., 1997; Thursby et al., 2009), suggesting aggressive commercial endeavors may lower the quality of academic research and serve as another source of negative effects on career development. Given these challenges, connecting with *strategic others* might help academic scientists to reap the opportunity and benefits of patenting with minimal costs. But who would be the *strategic others*?

According to social network theory, *trust* is contained in strong ties that facilitates the transfer of information and attitudes (Lin et al., 1981; Burt, 2000; Higgins and Kram, 2001), especially complex ones (Hansen, 1999). While collaboration is established and maintained by people who decide to jointly commit to a goal regardless of costs and difficulties, collaborative relationships certainly contain *trust*. Social network research also found relationships featured with *diversity* (ties across social divides) benefit individuals by bringing them additive information and opportunities as well as control power over scarce resources (Granovetter, 1973; Burt, 1992). Specifically in the scientific and technical context, scholars emphasize the benefits of social ties spanning organizational boundaries in transferring tacit knowledge and enhancing creativity (Bozeman et al., 2001; Dietz and Bozeman, 2005; Heinze and Bauer, 2007). Furthermore, researchers have demonstrated that the two features could work complementarily to promote the innovation process (Reagans and McEvily, 2003). As academic patenting could be considered as a boundary-spanning activity involving lots of learning, negotiation, and resource motivation in interaction with multiple levels of many organizations, the above points suggest academics' collaborators with distinct

background from them could be *strategic others* who are trustable and likely to provide novel information and resources to help academics surmount those challenges around patenting.

To what extent can we say a collaborator has a distinct background? Here I focus on their institutional affiliation.² Many studies have been conducted on the university-industry link and tend to find salutary effects on innovative activities (Audretsch and Stephan, 1996; Zucker and Darby, 1996; Zucker et al., 1998; Dietz and Bozeman, 2005). To my best knowledge, very few studies have tried to examine other cross-institutional relationships and their impacts on academic entrepreneurship. Only two studies attempted to examine the academia-government linkage (Strickland et al., 1996; Dietz and Bozeman, 2005), with the former indicating a positive effect and the latter a negative effect on academics' patenting engagement. Only one study touched interdisciplinary orientation and patenting activity and reports a strong association between them (Thursby and Thursby, 2011). Based on the institutional categories studied before, I decided to examine *collaboration ties with industry*, *collaboration ties with government*, and *collaboration ties with other disciplines*. Despite the absence of literature clues, I would also consider *collaboration ties with other universities* given patenting-related culture, knowledge, material and human resources may be unevenly distributed over universities and, once accessed, may change academics' minds and behaviors toward patenting engagement. All in all, I focus on *boundary-spanning collaboration ties* (the generic term of *collaboration ties with industry*, *collaboration ties with government*, *collaboration ties with other disciplines*, and *collaboration ties with other universities*) that contain both **trust** and **diversity** to investigate whether they are positively associated with patenting engagement and how they mediate the gender effects on patenting. Accordingly, the first hypothesis is meant to test the positive association between *boundary-spanning collaboration ties* and patenting engagement that is conceptually derived:

H1. All else being equal, *boundary-spanning collaboration ties* are associated with higher likelihood of patenting engagement.

As gender is focused upon, the social network research generally speaks to the differences between women and men in forming and using networks. In the professional labor force, women's networks tend to be small, dense, and intimate (Brewer and Lui, 1989), to have greater homogeneity (Ibarra, 1992, 1993), and to have contacts with less power and authority (Brass, 1985). The explanations for this pattern range from gendered socialization (Macdonald, 1992), workplace segregation (Brass, 1985; Griliches, 1990; Stanley, 1993; Dasgupta and David, 1994; Burt, 1998), dissimilar preferences for social activities in workplace (Podolny and Baron, 1997), to positional restrictions (Ridgeway, 1997). Those explanations are not exclusive but instead interrelated to each other and together reflect the widespread existence of gender beliefs and their powerful influence. However, we should note, regardless of their marginal status, women have made inroads into academic science and thus increased their share in the supply of potential collaborators within this profession. Indeed research (Bozeman and Gaughan, 2011) has evidenced that female academics tend to have more collaborators within academia. While the trend may be similar in all public sectors and Bozeman and Gaughan's finding may be extended to the

gender contrast in having academia-government ties, the stalled progress of women in top industrial management (Gupta et al., 2008; Tinkler et al., 2015) leads me to expect that collaboration with industry remains difficult for female academic scientists. Hence, the situation may be a bit complicated that requires specifications in the hypothesis:

H2a. Female academic scientists are not different from their male colleagues to have *boundary-spanning collaboration ties* in all types.

H2b. Female academic scientists are less likely than their male colleagues to have *boundary-spanning collaboration ties with industry*.

Research has explored the contingency of social network returns on gender. Overall, evidence is indicative of fewer social capital benefits for women even when they have similar networking behaviors (Brass, 1985; Ibarra, 1992; National Science Board, 2000; Campbell et al., 2002; van Emmerik, 2006). Lin (2001) used the concept *Return deficit* to explain the different outcomes by gender with similar networks. As women's limited experiences may be one reason for the underutilization of their strategic contacts, another possibility is that they feel uncomfortable (guilty or rude) to exploit important contacts (Walsh and Maloney, 2002). Taken together, Lin's *Return deficit* hypothesis can be extended to involve social processes in two scenarios: (1) women don't have critical contacts, and so even with mobilizing efforts, the returns are limited; (2) women have critical contacts, but (a) they are reluctant to mobilize, (b) they don't mobilize appropriately, (c) contacts are not willing to invest, and/or (d) the decision makers devalue women's social capital. However, there is a second possibility (Ding, 2004; Tinkler et al., 2015). While the benefits from having *boundary-spanning collaboration ties* may be smaller for women than men, considering women are usually excluded from information and resources exchanges, to the extent that they can access, these benefits are particularly valuable for them to promote outcomes. In other words, because women and men have different probabilities of accessing resources, the same amount of resources may have larger marginal impacts on women's professional outcomes than men's. Technically speaking, the two possibilities are that the interaction effect of gender and having *boundary-spanning ties* could be negative or positive, and corresponding hypotheses are:

H3a. Female academic scientists who have equivalent *boundary-spanning ties* are less likely than their male peers to engage in patenting activity.

H3b. Female academic scientists who have equivalent *boundary-spanning ties* are more likely than their male peers to engage in patenting activity.

Fig. 1 illustrates the conceptual framework that integrates the relationships between key and potentially relevant variables to explain women's lower level of patenting. The bold solid lines represent the hypothesized relationships to be tested. These are the direct effects of the various boundary-spanning collaboration relationships, and the indirect effect of being female through boundary-spanning collaboration on patenting performance. Relevant variables suggested by previous research, including boundary-spanning employment experiences (Dietz and Bozeman, 2005; Melkers and Xiao, 2012; Scellato et al., 2014), work context (Agrawal and Henderson, 2002; Ding et al., 2006; Azoulay et al., 2007), academic rank (Mowery, 2007), scientific achievement (Kira and Alberto, 2005; Thursby and Thursby, 2005; Azoulay et al., 2007; Breschi et al., 2007; Whittington and Smith-Doerr, 2008), are specified as controls to assure comparisons are made between scientists who have similar resources and standings.

² There are two reasons for my focus on institutional affiliations rather than geographic locations. First, the contemporary transportation and ICT development significantly lower the barriers for information and resource exchange. Second and more importantly, I conceive of the heterogeneity in information, resources, norms, and practices is more prominent across institutional settings and more influential to the motivations and behaviors toward commercial engagement. The second reason is also why I did not consider demographic differences in this study such as those in gender, race, and age.

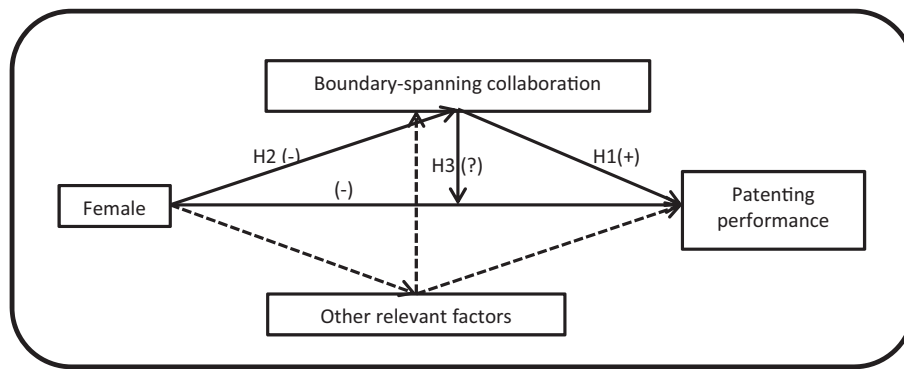


Fig. 1. Overall theoretical framework.

3. Data and method

3.1. Data source and manipulation

The data used for this study are drawn from “Women in Science and Engineering Network Access, Participation, and Career Outcomes” (NETWISE), a major national study of academic scientists in the U.S. In NETWISE, a survey was conducted of 3677 individuals randomly drawn from the population of academic scientists and engineers in six disciplines (biology, chemistry, physics, computer science, earth and atmospheric sciences/EAS, and electronic engineering) in 150 U.S. Research I universities. The sample was stratified by gender, rank, and discipline.³ The initial invitation and follow-up reminders were sent to individuals in the sample by traditional mail, but the survey was implemented online using Sawtooth Software. 1598 usable responses were finally received (44% response rate). Responses were fairly evenly distributed across the six disciplines (around 18% from EAS, 17% from biology, chemistry, and physics, 16% from computer sciences, and 13% from electronic engineering) and genders (48% women) but a little skewed toward senior ranks (27% assistant professor, 28% associate professor, and 45% full professor).

This NETWISE database is unique because, in addition to academic scientists’ demographic information, research activities, publication productivity, teaching and administrative responsibilities, and perception on institutional environment, it contains the background of their collaborators and the relationships with collaborators (e.g. origin of acquaintance, closeness of research expertise, communication frequency, the type of collaboration, etc.). However, the survey was not intended to study scientists’ patenting activities and then the patenting performance information was not originally available. But fortunately, with detailed identification data available, I was able to search and collect patent data from the USPTO patent database. To minimize the problem with endogeneity, I manipulated collaboration as an exogenous variable, that is, I purposively searched patents filed after the collaboration ties were established (indicated in the 2006 survey) in 2006–2010.⁴ As the 5-year window is relatively short for sufficient cases to have patent applications to be granted (given the thorough substantial examination⁵ that spans at least 12 months), I decided to use patent

applications (rather than patent grants) to increase the instances that had ever filed a patent in this sample.

To search and add additional patent information to the survey data, the survey respondents were first verified if they remain active in the US academic sector. Those who died, moved to other employment sectors, or left the US were dropped. For each of the active academic scientists in the survey, their patent application information was manually searched in the USPTO online patent database. After the search was completed and information was recorded, two rounds of verification were then conducted to ensure such information is precise and reliable. After correction and further confirmation, 258 scientists, or 16.4% of the sample, were identified to file at least one patent application in the 5-year period of study. Looking at the faculty scientists’ rate of patenting by discipline (Table 1), I found the rate of patenting involvement in the EAS group is too small to indicate EAS research is likely to be commercialized, and so excluded the EAS scientists ($N=288$) for the following analyses. Ultimately, the study sample consists of 1283 individual scientists, among which 252 (or 19.6%) had patent application(s) in the 2006–2010 period. It is ideal to test if there is any self-selection bias among the survey respondents. While this could be done by examining whether the gender contrast of ever-patenting ratios is comparable between respondents and non-respondents, it is impossible in practice as searching the patent information of faculty members in the original sample requires far more time and resources outside my control. As a reasonable alternative, however, the ratios derived from Blume-Kohout’s study (Blume-Kohout, 2014), 12.1 among females and 27.4 among males, are drawn for the comparison. Apparently, some bias exists in the sample of current study (17.7 among females and 21.6 among males, indicated in Table 3), indicating female faculty who have ever patented tended to respond the survey but male faculty who ever patented tend to not respond the survey. This bias nevertheless does not significantly distort the conclusions of this study but is discussed when attempts are made of generalizing the findings.

Table 1

The distribution of patenting scientists (2006–2010) by discipline.

	Biology	Chemistry	CS	EAS	EE	Physics	Total
Total N	275	275	253	288	207	273	1571*
Patented (%)	10.2	27.6	20.2	2.1	32.4	11.0	16.4

Note: CS, computer science; EAS, earth and atmospheric sciences; EE, electronic engineering.

* While there were 1598 valid cases in the original dataset, 27 were found inactive in the U.S. academia for various reasons during the studied period (2006–2010) when collecting patent data and thus excluded.

³ The selection of these disciplines was based on the level of women’s representation, that is, disciplines with low, transitioning, and high level of women’s representation.

⁴ 2010 is the most recent year that the whole year patent data were available when the current research was conducted.

⁵ <http://www.wipo.int/sme/en/faq/pat/faqs.q4.html> and http://www.uspto.gov/patents/resources/general_info_concerning_patents.jsp.

Table 2
Variables and their definitions.

Variables	Definition
<i>Dependent variable</i>	
Patenting involvement	A dummy variable with 1 indicating a scientist filed at least one patent application during the five-year period 2006–2010 (the dependent variable in the multiple logit analysis).
Patenting productivity	An interval variable indicating the total counts of an academic scientist's patent applications in 2006–2010 (only included in descriptive analysis).
<i>Independent variable</i>	
Female	A dummy variable with 1 indicating a woman scientist.
Collaboration network size	A interval variable indicating the total number of collaborators an academic scientist has.
Industrial ties	A dummy variable indicating whether an academic scientist has at least one collaborator in industry.
Governmental ties	A dummy variable indicating whether an academic scientist has at least one collaborator in government.
Inter-university ties	A dummy variable indicating whether an academic scientist has at least one collaborator in other universities.
Interdisciplinary ties	A dummy variable indicating whether an academic scientist has at least one collaborator in other departments.
<i>Control variable</i>	
Prior industry employment	Dummy variable indicating whether an academic scientist had full-time employment in industry before.
Prior government employment	Dummy variable indicating whether an academic scientist had full-time employment in a government agency or a NPO before.
Postdoctoral appointment	Dummy variable indicating whether an academic scientist had postdoctoral appointment(s) before.
Joint appointment with other department(s)	Dummy variable indicating whether an academic scientist had formal joint appointment with another academic department.
Lab/Center affiliation	Dummy variable indicating whether an academic scientist had membership with a permanent S&E lab or center.
# of years since Ph.D.	Interval variable indicating the number of years from an academic scientist earned his/her Ph.D. to 2006.
Academic leadership administrative position	Dummy variable indicating whether an academic scientist had a high-profile administrative position.
# of journal articles	Ordinal variable indicating the number of journal articles that an academic scientist published in previous two years.
# of conference proceedings	Ordinal variable indicating the number of conference proceedings that an academic scientist published in previous two years.
Grant amount	Interval variable indicating the total amount of grants awarded to an academic scientist in the previous two years (its natural logarithm is used in multiple logit analysis).
# of Ph.D. students supervised	Interval variable indicating the number of Ph.D.s an academic scientist supervised in the previous five years.
Biology	Dummy variables indicating an academic scientist's disciplinary affiliation with Physics as the reference group.
Electronic engineering	
Chemistry	
Computer sciences	
Active TTO in R's university(2005)	Dummy variable indicating whether an academic scientist's employing university had an active TTO in 2005.
The patent application stock of R's university(2005)	Interval variable indicating the number of patent applications of an academic scientist's employing university in 2005.

3.2. Key variables

3.2.1. Patenting involvement

With the patent application information available, one dummy variable was created to indicate whether a scientist filed at least one patent application during the five-year period 2006–2010. This will be the dependent variable in the multivariate regression analyses to test the hypotheses about the effects (through collaboration) of gender. A numeric variable, patenting productivity, was also created to indicate patenting performance in terms of the total number of patent applications filed in the studied period. But the latter variable is only used in the descriptive analyses.

3.2.2. Gender and collaboration network

The primary analytical variables in the models are respondents' sex (*Female*) and the four specified types of *boundary-spanning collaboration ties*. 7876 individuals were named as close collaborators by the survey respondents through a few questions about "closest research collaborators" where "collaboration includes proposal generation, working on a research project, writing/presenting an academic paper/book or book chapter, or developing industrial products or patents." That is, those collaborators not only worked together with the focal academics but also made substantial contributions to their research. In this sense, to the academics, these reported collaborators are highly trustable others. According to these collaborators' organizational affiliation, four dummy variables were created to indicate whether an academic scientist investigated had one or more collaboration ties with industry (*Industrial ties*), with government or public agency (*Governmental*

ties), with other universities including foreign universities (*Inter-University ties*), and with other departments (*Interdisciplinary ties*). Because previous research suggests that network size is a proxy of social capital of diversity, I also include the total count of each academic scientist's collaborators (*Collaboration network size*)⁶ to clearly sort out the effects of social ties' type from size. Moreover, interaction terms were created to test whether the effects of the collaboration network variables have differentiated effects on patenting involvement by gender as hypothesized.

Other possible variables indicating academic scientists' experiences, positions, scientific achievement, and organizational environment, were defined and introduced in the multivariate regressions to single out the net effects of the key variables on patenting involvement. All the variables initially included for this investigation, as well as their definition, are listed in Table 2.

3.3. Analytical method

I first gauged the gap between genders on patenting performance. The means of the female and male group on patenting performance measures, *Patenting Involvement* and *Patenting Productivity*, were compared. Then, I compared the means between genders on all the boundary-spanning collaboration tie variables,

⁶ The squared term of this measure was created and tested for model building, but there was little significant difference between results from models with and without the square term. Thus, in the final report, I do not include it in the models.

Collaboration network size, Industrial ties, Governmental ties, Inter-university ties, and Interdisciplinary ties.

The third step was to test the hypotheses concerning how *boundary-spanning collaboration ties* interact with gender to affect patenting engagement. Multivariable analyses were adopted on this stage. The goal here is twofold, to estimate whether *boundary-spanning collaboration ties* are strong predictors of patenting involvement and whether and how the introduction of them can help explain the gender difference in patenting involvement. *Patenting involvement* was the dependent variable used in the analysis and given the dichotomous nature of this dependent variable, logit regression were used for model estimation. The multivariate logit analysis allows variables to enter the equation in groups. By introducing variables in group, I could tease out the net effect of the key collaboration variables' effects and also their interaction effects with gender. It has to be noted that, regardless of the long list of control variable initially considered, the final models reported here only contain the control variables that are significantly explain the variance in the dependent variable. The significance is determined by χ^2 statistics of the model after introducing each factor. The multivariate analysis design can be expressed as the following equation:

Odds of patenting involvement

$$= f(\text{female} + \text{boundary-spanning collaboration} + \text{female} \\ \times \text{boundary-spanning collaboration} + \text{other relevant factors})$$

4. Analysis and findings

The following analyses are restrained to 1261 cases that have valid value on all the selected variables. Among these academic scientists, 45.3% are female. The similar proportion is due to our purposive sampling method. In addition, although more males are tenured than females (χ^2 test $p < 0.05$), both genders have similar distribution over the ranks: 29, 28, and 43% of the females are assistant, associate, and full professors (compared to 26, 28, and 46% in the male group). The female scientists tend to be younger than their male colleagues, with the mean age of the former group being 45.9 and the latter 47.6 (one-tailed t -test $p < 0.01$). The distribution of scientists across the five disciplines is pretty balanced (around 20%), and is not significantly different by gender.

Table 3 presents the descriptive results of patenting involvement and productivity by gender in our sample. A gender gap is clearly shown, but the gap only exists when taking everyone into account but not among those who have ever patented. In other words, the gender difference primarily lies on the “involvement” stage and women's disadvantage in patenting rate seems negligible when only patenting scientists are focused on. Considering a self-selection bias exists in the current study sample, it is likely that the gap is larger in the population. But the results about the

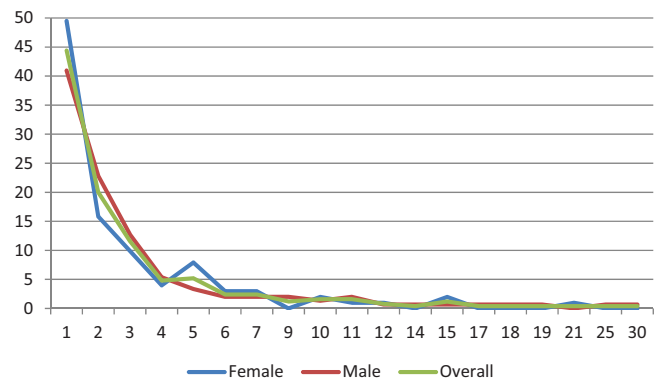


Fig. 2. The proportion of patenting scientists by the number of patent applications.

significant difference in ever-patenting ratio and the insignificant difference in actual patent counts are the same as those reported by Blume-Kohout (2014). As such, patenting involvement rather than patenting productivity will be the focus for our next exploratory steps. But it is still informative to look at patenting scientists and their productivity by gender. As Fig. 2 shows, the distribution on patenting productivity was highly skewed. In particular, those who had filed more than five patents only accounted for 14% in the sample, but they produced almost half of the patent applications ($n = 813$). The three lines in the figure are so close, suggesting no prominent gender distinction among the patenting scientists.

I then moved to examine the academic scientists' collaboration patterns (Table 4). First, the result shows these scientists' collaboration ties ranged from 0 to 10, with the average being 5, a reasonable number suggested by previous research (Marsden, 1987; McPherson et al., 1992; Walsh and Maloney, 2002). Second, a significant majority of these academic scientists (71.4%) had collaborators in other university, suggesting inter-university collaboration has been commonplace among academics, at least in the studied disciplines and research intensive universities. Interdisciplinary collaboration within the same university ranks the second in popularity. However, only a small share of academic scientists had built cross-sector collaborations, especially academia-industry collaborations. More relevant to our focus is that women seemingly achieved parity with their male colleagues in terms of building boundary-spanning collaboration. There is no statistically significant difference between female and male scientists on these boundary-spanning collaboration measures, and in fact females are even more likely to have collaborative connections with government or public agencies than their male peers. The results support H2a but not H2b, though in this sample female scientists tend to have less collaboration ties with industry than male scientists.

To test the power of *boundary-spanning collaboration ties* in explaining the gendered pattern in patenting involvement, I regressed *Patenting involvement* against the five measures of boundary-spanning collaboration and their interactions with

Table 3
Comparing female and male scientists on patenting performance.

	Overall (N = 1261)	Female (N = 571)	Male (N = 690)	GD <i>t</i> -test
Patenting involvement %	19.8	17.7	21.6	−3.9*
Patent productivity				
Mean (including non-patenting scientists)	0.64 (2.24)	0.53 (1.84)	0.74 (2.52)	−0.21*
Mean (including only patenting scientists)	3.25 (4.10)	2.98 (3.44)	3.44 (4.49)	−0.46

Note: For the mean comparison in the 3rd row, only 250 cases in total, 101 female, and 149 male, who have ever patented are included. GD, gender difference; standard deviations are in parentheses; One-tail t -test was applied.

* $p < 0.05$.

Table 4

Comparing scientists on boundary-spanning collaboration by gender.

	All (N = 1261)	Female (N = 571)	Male (N = 690)	GD <i>t</i> -test
Collaboration network size	4.38 (2.80)	4.50 (2.86)	4.28 (2.73)	0.22
Industrial ties (%)	10.8	9.8	11.6	−1.8
Governmental ties (%)	19.2	22.4	16.5	5.9**
Inter-university ties (%)	71.4	72.3	70.6	1.7
Interdisciplinary ties (%)	41.6	43.1	40.4	2.6

Note: GD, gender difference; standard deviations are in parentheses; one-tail *t*-test was applied.** $p < 0.01$.

gender, controlling for a series of selected variables. As stated before, the results presented here only concerned with the factors showing a significant association with the dependent variable in addition to the key independent variables (gender and boundary-spanning collaboration). The correlation matrix of all variables included in the final models shown in Table 5 provides the Pearson's *r* of each pair of these variables. The matrix suggests most of the variables included are correlated with each other but none of the Pearson's *rs* have the value higher than 0.40, suggesting the multicollinearity problem not be an issue for the following multivariate analyses.

Table 6 presents the results of six logit regression models, with the first four covering all cases in our sample and the last two for female and male academic scientists respectively. Wald test results are reported as well and show that all models are significant at the 0.001 level except Model 1 that is significant at the 0.01 level. Unless noted specifically, all important estimates are compared and discussed below as all other variables are held at their means. Across the first three models, the odds ratios and significant test results of *Female* suggest the gender gap remained pronounced even after adding controls and boundary-spanning collaboration variables. Model 1 indicates that women were less likely to engage in patenting than men, with their odds of engagement being about 63% as high as men's. The difference is statistically significant, which is consistent with the result from comparison of means between genders on patenting involvement.

Control variables remained significant in our trial efforts were included in Model 2. They include full-time employment experiences in industry, having joint appointment with other department(s), working in the field of chemistry (instead of physics, biology, electronic engineering, or computer sciences), Ph.D. cohort, the number of conference proceedings, and the number of supervised Ph.D. students. The inclusion of these variables significantly increase the power of the model in explaining the variance in patenting involvement as the Wald χ^2 statistics increases by 115.49 and significant at the 0.001 level with 6 degrees of freedom.

Moving to Models 3 and 4, we found the simple introduction of boundary-spanning collaboration variables would not enhance the explanatory power of the model (the Wald test result suggests these variables do not significantly increase the Wald χ^2 statistics), but including the combination of these variables and their interaction terms with gender would do. In addition, in Model 4 the odds ratio of *Female*, while remains smaller than 1, is not statistically significant any more, suggesting the specification of boundary-spanning collaboration's indirect effects would be appropriate to understand the gender gap in patenting involvement. While it is always tricky to interpret the interaction terms, we ran Model 3 using the female and male subsamples separately (Models 5 and 6). The results show, despite almost all specified boundary-spanning collaboration ties tend to increase patenting involvement, only *Industrial ties* is the strong predictor of such involvement among female scientists.

A better understanding of the results could be developed in combination with other factors. Previous research has suggested that male faculty scientists have more exposure to commercial processes through either invited conferences involving industry participants or consultancy provided by previous students working in industry (Ding et al., 2006; Murray and Graham, 2007; Stephan and El-Ganainy, 2007). Our results from Models 5 and 6 offer the support to this point as they show that, among male academic scientists, the number of conference proceedings and previous doctoral students are strongly associated with higher probability of patenting involvement. Meanwhile, in the female group, the most influential factors are full-time employment in industry and collaboration with industry, implying the commercialization related knowledge and resources are quite restricted for women to fewer channels where a higher level of engagement and trust is required. To conclude, the results suggest that collaboration with industry could significantly increase the likelihood of patenting engagement for female academics but this is not the case for male academics. In other words, from the perspective of women, H1 and H3b seem to be acceptable. However, the interpretation is not fully warranted given the existence of bias in the study sample.

Table 5

Correlation matrix of variables used in the logit regression analyses.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
(1) Patent involvement	1												
(2) Female	−0.049*	1											
(3) Collaboration network size	0.070*	0.040	1										
(4) Industrial ties	0.129*	−0.029	0.220*	1									
(5) Governmental ties	−0.005	0.075*	0.297*	−0.033	1								
(6) Inter-university ties	0.038	0.019	0.610*	0.051	0.086*	1							
(7) Interdisciplinary ties	0.109*	0.027	0.395*	0.080*	0.091*	0.172*	1						
(8) Prior industry employment	0.148*	−0.013	0.089*	0.143*	−0.018	−0.007	0.032	1					
(9) Joint appointment with other department(s)	0.082*	0.051	0.162*	−0.005	0.004	0.109*	0.154*	0.010	1				
(10) # of Years since Ph.D.	−0.065*	−0.091*	−0.103*	−0.033	−0.016	−0.064*	−0.061*	−0.185*	0.046	1			
(11) # of conference proceedings	0.187*	0.004	0.162*	0.203*	−0.020	0.057*	0.010	0.216*	0.079*	−0.175*	1		
(12) # of previous Ph.D. students	0.252*	−0.023	0.119*	0.117*	−0.011	0.065*	0.081*	−0.090*	0.109*	0.261*	0.187*	1	
(13) Chemistry	0.109*	0.038	−0.084*	−0.057*	−0.018	−0.033	0.089*	−0.081*	−0.052	0.027	−0.335*	0.104*	1

* $p < 0.05$.

Table 6
Logit models predicting Patenting Involvement (robust standard errors are in parentheses).

	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Prior industry employment		2.14** (0.63)	2.27** (0.69)	2.23** (0.68)	4.38*** (1.75)	2.03* (0.70)
Joint appointment with other department(s)		1.99** (0.49)	1.89* (0.48)	1.89* (0.49)	1.25 (0.42)	1.99* (0.57)
# of years since Ph.D.		0.97** (0.01)	0.97** (0.01)	0.97** (0.01)	1.00 (0.02)	0.97** (0.01)
# of conference proceedings		1.26*** (0.07)	1.26*** (0.07)	1.26*** (0.07)	1.20* (0.09)	1.27*** (0.08)
# of Ph.D. students supervised		1.29*** (0.05)	1.29*** (0.05)	1.29*** (0.05)	1.21** (0.07)	1.30*** (0.06)
Chemistry		3.23*** (0.78)	3.01*** (0.74)	3.02*** (0.74)	2.57** (0.84)	3.12*** (0.87)
Female	0.63** (0.10)	0.65* (0.11)	0.65* (0.11)	0.76 (0.29)		
Collaboration network size			0.93 (0.05)	0.93 (0.05)	0.98 (0.08)	0.93 (0.06)
Industrial ties			1.25 (0.38)	1.08 (0.37)	3.37** (1.27)	1.08 (0.38)
Governmental ties			1.12 (0.30)	1.06 (0.33)	1.37 (0.47)	1.06 (0.33)
Inter-university ties			1.57 (0.42)	1.64 (0.49)	0.84 (0.32)	1.63 (0.49)
Interdisciplinary ties			1.53 (0.34)	1.55 (0.38)	1.48 (0.48)	1.53 (0.38)
Female × Collaboration network size				1.02 (0.10)		
Female × Industrial ties				3.75* (1.96)		
Female × Governmental ties				1.40 (0.65)		
Female × Inter-university ties				0.51 (0.25)		
Female × Interdisciplinary ties				0.97 (0.41)		
Constant	0.27*** (0.03)	0.09*** (0.02)	0.07*** (0.02)	0.07*** (0.02)	0.04*** (0.02)	0.07*** (0.03)
Wald χ^2	8.65**	117.58***	122.63***	137.27***	67.53***	98.58***
Observations	1261	1261	1261	1261	571	690

* $p < 0.10$.

** $p < 0.05$.

*** $p < 0.01$.

Although I speculate that the differential effects of industrial ties might be even more prominent when more ever-patenting male academics were included in the analysis, further investigation and verification is needed.

5. Discussion and conclusion

The rapid increase of academic scientists' involvement in various commercial activities suggests the arrival of entrepreneurial regime. Concerned with women's situation in the new trend, this study sfocuses on collaboration as the potential social mechanism to inquire how academic patenting is gendered. Drawing upon social capital theory, I conceptualize the importance of academic scientists' social ties featured with **trust** and **diversity** to their patenting engagement, and develop the concept *boundary-spanning collaboration ties* and three hypotheses around this concept for empirical testing.

The major findings can be revisited in relation to each of these hypotheses. First, female academic scientists are not disadvantaged in collaboration, whether in terms of general collaboration network size or having any of the specified *boundary-spanning collaboration ties*. While the significance test results only supports H2a, the gender contrast with women having less collaboration ties with industry in the sample is consistent with what H2b hypothesized. Second, with respect to the relationship between

boundary-spanning collaboration ties and patenting involvement, the results suggest that there is no relationship between the two for men in statistical sense; but for women, having collaboration ties with industry would significantly promote their engagement in patenting. Furthermore, the differential effects of collaboration ties with industry to a large extent help explain the gender distinction in patenting engagement. Again, from women's lens, we could say the evidence support H1 and H3b.

Some further thoughts could be drawn from the results. First, it seems the critical resources to promote academics' patenting engagement reside in the social ties with industry. They could be the idea about the potential market, the knowledge about the current industrial production process and the actors in the value chain, the acquaintance of the capabilities of available engineers and technicians, additional financial sources (e.g. venture capital and bank loans), among many other things. Second, while these resources are important to academics in both sex groups, they could be accessed by male academics through a broad range of avenues, especially those *weak ties* involving a lower level of **trust** (e.g. conference participation and student contacts). By contrast, they tend to be available for female academics through restricted channels requiring a higher level of **trust**. Third, the model predicts that the critical resources, once accessed, would significantly promote women's engagement in patenting. In this sense, Lin's (2001) *return deficit* hypothesis is not supported. Instead, the implication is that women's credential and capabilities could be appropriately judged

by *strategic others* once *trust* is built between them and the *strategic others* and then could leverage the needed resources for their better outcomes. This finding and interpretation is consistent with other scholars' (Burt, 1998; Ding et al., 2006; Tinkler et al., 2015).

The findings and interpretations, with limitations in mind, should be viewed as worthy directions for future exploration rather than conclusive assertions. A major limitation lies in the lack of qualitative information that would have deepened the understanding of the revealed patterns in this study. Without evidence from in-depth interviews or case studies, explanations could not be offered on why some conceptually relevant factors (e.g. the number of journal article publications, the amount of research grant, the university environment for commercialization measured through patent applications at the university level) have no impacts on patenting involvement. Neither could it be totally confirmed that women receive little commercial activity related resources through ways other than collaboration with or employment in industry.

Secondly, the dependent variable is rather simple. Despite the focus on "involvement" rather than "productivity" might be appropriate considering the ever-patenting ratio by gender is sharply contrast here and in other studies (Whittington and Smith-Doerr, 2005; Blume-Kohout, 2014), more measurements of patenting performance (e.g. the forward citations of patents) should be examined to enrich our understanding of the gender gap in patenting, especially from the perspective of collaboration ties. Thirdly, the study sample is restrained to tenure-tracked faculty in research intensive universities, leaving other kinds of higher education institutions and non-tenure tracked researchers unexamined. This limitation is especially unfavorable to research on gender and patenting because evidence both suggests that women are over represented in less prestigious universities and unstable positions (Long and Fox, 1995) and that patents from higher education institutions other than Research I universities and non-tenure tracked scientists account for a sizable share (Azoulay et al., 2007; Blume-Kohout, 2014). Inclusion of individuals representative of all groups in higher education is then expected to help justify and generalize the current results. A third limitation is the short time frame (5 years) for patent application data collection. Despite the deliberate differentiation of the time for scientists' patent data and their collaborative activities, the relatively short time window retains the difficulty in completely sorting out motivations from the effects of collaboration. In other words, it is possible that building collaborative relationships is motivated by patenting. Then a time lag analysis design for longitudinal data could better address the causal relationships. Another limitation is the possible self-selection bias introduced in the survey process. Such a bias could be detected and removed in the future research with specific thought and resources devoted to this aspect.

Regardless of these limitations, this study generates new knowledge in the broad research area of gender inequalities in science and sheds light on possible policy intervention. Based on the findings presented in the paper, I would argue: on the one hand, female academic scientists are better off on some formal behavioral measures (like collaboration), which tends to result from many preferential programs and policies aiming to promote gender equality; and on the other hand, the influence of gender beliefs remain quite strong so that important resources for professional performance are restricted for women to channels a high level of *trust* is required. While consensus has not been reached about whether entrepreneurial activities should be encouraged in academia, specific policy recommendations may not be given at this point as to whether and how to promote women's participation in patenting (and other commercial activities). Instead, an appropriate recommendation could be two-step action: First, support the gender studies of science so as to accumulate knowledge about scientists' attitudes, perceptions, and actual behaviors regarding academic

patenting and other forms of academic entrepreneurship; second, ground policy agendas, formation, and implementation firmly on the knowledge base and develop an integrative policy framework. It should also be expected that the accumulation and publicity of the research could help to overturn the fundamental culture against women in science.

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